

A Miniaturized WBAN-Compatible Parasitic Patch Antenna for Non-invasive Brain Tumor Localization

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Abstract—This study introduces a miniaturized wideband patch antenna design, specifically developed for electromagnetic biomedical applications. The antenna (25 mm×25 mm) operates within a bandwidth range of 6.68 to 8.82 GHz, demonstrating a stable radiation pattern with a low specific absorption rate (SAR) (<1.6 W/Kg). It excels in delivering high-resolution imaging and facilitating deep tissue penetration. The geometry of the antenna is optimized to achieve dual functionality, enabling its application in both brain tumor detection and IEEE 802.15.6 WBAN healthcare communication. On a simulation platform, to avoid false negative results, the tumor was placed at different coordinates inside the phantom and verified the presence of malignant tumor by changing in reflection coefficient and other radiation parameter with respect to healthy tissues.

Keywords— Brain tumor detection, WBAN, wideband antenna, SAR, biomedical applications, phantom.

I. INTRODUCTION

In recent years, the exploration of innovative antenna designs has become imperative, especially in the realm of biomedical applications. Miniaturized antennas with low SAR enable non-invasive microwave imaging, a cost-effective alternative to ultrasound, CT, MRI, and radiography [1]. Tumors, whether primary or metastatic, benign or malignant, pose serious health risks, making early detection crucial for effective treatment. In 2020, over 251,000 global deaths were attributed to early-stage brain tumors [1], and cancer incidence and fatalities are projected to rise sharply by 2040 projected to report 2,001,140 new cancer cases and 611,720 deaths [2]. Ultra-wideband (UWB) antennas (3.1–10.6 GHz) offer high penetration, accuracy, and low-power operation, suitable for microwave imaging and healthcare applications such as IEEE 802.15.6 WBANs. Various antenna types including microwave, UWB, resonant (DRA, loop), millimeter-wave, PIFA, Vivaldi, and metamaterial antennas are employed to exploit dielectric contrasts between

healthy and tumor tissues for anomaly detection [3]. In [5-15] papers of various antenna designs and their utilization in detecting brain tumors has been analyzed and their comparison is shown in Table IV. A Hilbert curve patch antenna on a 57.6 mm square substrate achieved minimal SAR (0.162 W/kg) [4], while a DGS-based UWB monopole antenna operated effectively at 2.94–6.96 GHz [5]. M. Samsuzzaman et al. proposed a circular-slot antenna (60×70 mm²) with >5 dBi gain and 85% efficiency [6], and a W-slot U-shaped patch antenna achieved high directionality (>3.5 dBi) for MWI [7]. Similarly, an octagonal slotted patch antenna (43×40 mm²) showed -50.55 dB return loss and SAR within safety limits [8]. The study present in [9] is designed compact microstrip patch antennas offer UWB operation with gains ~5.8 dB but moderate power handling. Designs based on Rogers RT5880 substrates showed SAR 0.34–0.63 W/kg with -57.98 dB return loss proposed by I. Abtahi et al. [10]. A slotted patch antenna with DGS achieved effective tumor differentiation with -25.49 dB reflection coefficient offered in [11]. Another design by A. J. D et al. developed a pentagon-shaped antenna with -22 dB return loss at 5.8 GHz [12], a rectangular patch antenna in 5G band achieving low SAR value of 0.000635 W/kg SAR [13], and a compact monopole antenna detecting 5 mm tumors with 84 MHz frequency shifts in [14]. However traditional microwave antennas face challenges like large size [5-9], [12-15], and narrow bandwidth [5], [8] and [13] which restricts resolution and reduces the ability to detect small tumors or achieve accurate imaging. Patch-based slotted designs are easy to fabricate, low-cost, and suitable for integration into portable or wearable tumor detection systems. In this context, the present paper introduces a miniaturized UWB antenna (25×25×0.79 mm³, 6.68–8.82 GHz) with improved efficiency, resolution, and safety limit. Section II covers the design; Section III discusses antenna performance analysis in free space and explores the tumor detection process. Section

IV compares with recent work progress in this field and finally in section V concludes with a critical evaluation.

II. ANTENNA STRUCTURE

UWB microstrip patch antennas are well suited for microwave tumor imaging due to their wide bandwidth, low return loss, and strong tissue penetration. Fig. 1 presents a compact patch antenna designed on a Rogers RT5880 substrate ($\epsilon_r = 2.3$, $\tan\delta = 0.0009$, thickness = 0.79 mm) with overall dimensions of 25 mm $\times$ 25 mm. The ground plane (19 mm $\times$ 5 mm) is optimized to enhance bandwidth. A 7 mm $\times$ 2.3 mm microstrip feedline provides 50 Ω impedance matching. The design integrates a 14 $\times$ 1.30 mm² parasitic strip and a 0.9 mm circular ring to achieve broadband operation from 6.68–8.82 GHz with improved gain and efficiency. Detailed design steps, physical dimensions, governing equations, and free-space performance analysis are provided in [16].

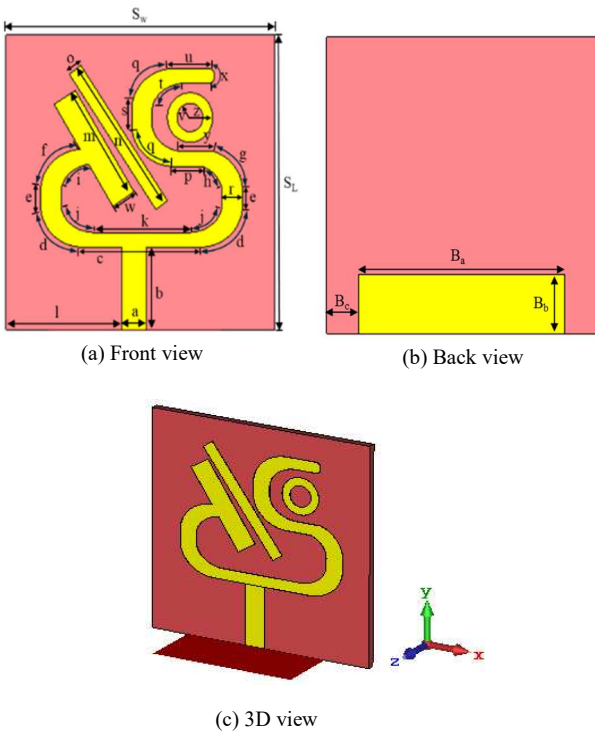


Fig. 1. Proposed patch antenna [16].

III. PERFORMANCE ANALYSIS

Designing an antenna for a human head model requires consideration of placement, size, frequency range, and body effects, with performance evaluated via return loss, VSWR, SAR, and electric field intensity. The Hugo voxel head model (35-year-old male) includes six primary brain tissue layers skin, fat, cortical bone, CSF, white matter, and grey matter (Fig. 2) along with surrounding tissues such as blood, cartilage, cerebellum, eyes, muscles, and glands, each with frequency-dependent permittivity, conductivity, and density (Table I). Simulation in CST Microwave Studio with the antenna positioned 20 mm from a healthy head shows a reflection coefficient of -13.66 dB at 7.79 GHz in Fig. 3. Introducing a 5 mm tumor ($\epsilon=67$, $\sigma=7$ S/m, $\rho=1000$ kg/m³, $k=0.56$ W/K $\cdot$ m) shifts S_{11} to -15.99 dB and resonance to 7.87 GHz (6.68–8.82 GHz). A small shift in the resonance frequency indicates that the antenna is responsive to changes in the tissue environment, which occurs in the presence of a

tumor with distinct dielectric properties. More significantly, variations in the return loss ($|S_{11}|$) provide a clearer indication of how the tumor affects the antenna impedance and the surrounding electromagnetic fields, making it a more robust and reliable marker. Tumors typically exhibit electrical properties that differ from those of healthy tissue, which can result in increased reflection of electromagnetic waves at the tumor boundary and alterations in the electromagnetic characteristics of the tissue. These effects can cause impedance mismatches between the antenna and the tissue, leading to an increase in the voltage standing wave ratio (VSWR). Consequently, more energy is reflected toward the antenna rather than being absorbed by the tissue. As shown in Fig. 4, the VSWR changes from 1.4 to 1.55, reflecting this behavior. The presence of a tumor can cause changes in the directivity pattern of the antenna with alternation in the main lobe direction, shape and strength that captured in Fig. 5 (a, b) changes from 1.82 to 2.58 dB. In the presence of a tumor (in Fig. 5 b), the directivity increases because the malignant tissue absorbs and distorts the radiated field, reducing the antenna's ability to focus energy in a specific direction.

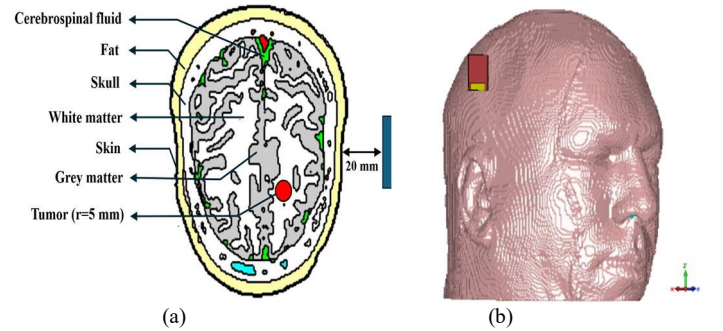


Fig. 2. (a) Cross-sectional view of human head phantom with tumor and (b) 3D simulation setup.

TABLE I BRAIN TISSUES PROPERTIES [7]

Tissue	Relative Permittivity	Conductivity (S/m)	Density (Kg/m ³)
CSF	70.1	2.3	1007
Fat	5.54	0.04	812
Skull	12.45	0.24	1800
White matter	43.22	1.01	1041
Skin	45	0.73	1100
Grey matter	46	0.9	1174
Tumor	67	7	1000

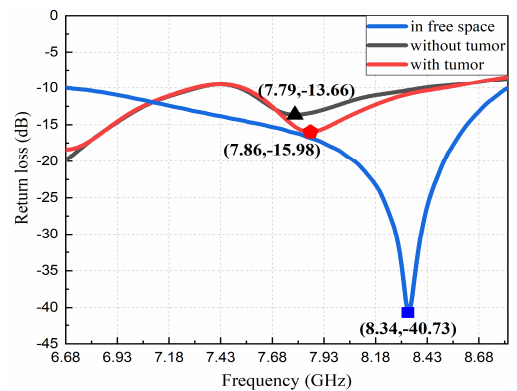


Fig. 3. $|S_{11}|$ analysis in free space, without tumor and with tumor condition.

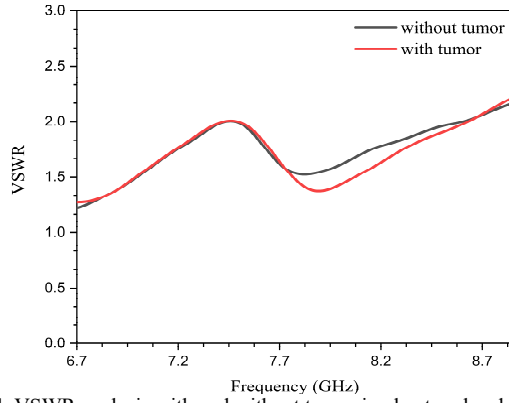


Fig. 4. VSWR analysis with and without tumor in phantom head model.

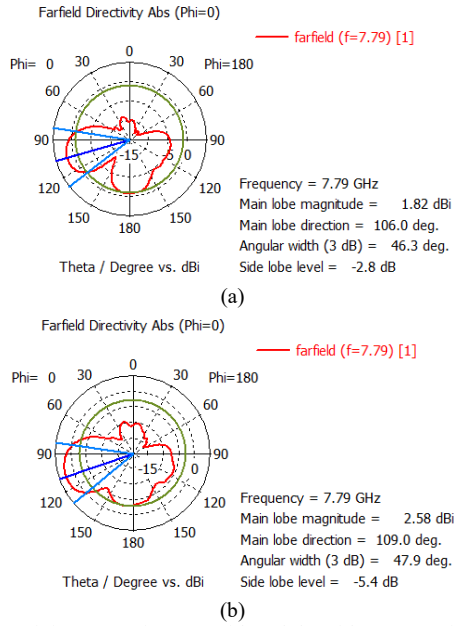


Fig. 5. Directivity comparison (a) with and (b) without tumor in phantom head model.

Tumors alter the antenna radiation pattern, as illustrated in Fig. 6(a), the maximum E-field in a head without a tumor was recorded as 30223.7 V/m at a frequency of 7.79 GHz. However, in the presence of a tumor on the head, the maximum E-field decreased to 28602.7 V/m at 7.79 GHz frequency (Fig. 6b). This decrease can be attributed to the increase in water content within the tumor compared to the surrounding healthy brain tissue, which affects the overall electrical capacity of the tissue, thereby reducing the E-field. Fig. 7 indicates that the values of current density can increase maximum 198.35 to 247 A/m² with the presence of a tumor. Tumors absorb more energy from electromagnetic radiation compared to surrounding healthy tissue due to their altered properties. This lead to higher SAR values compared to healthy head model shown in Fig. 8 (a, b) in the tumor region where max SAR value noted 1.37 W/Kg at 7.79 GHz frequency which is in recommended range (<1.6 W/Kg) proposed from FCC. That quantifies the amount of electromagnetic energy absorbed by biological tissues and is essential to ensure patient safety during diagnostic exposure, particularly when the antenna is placed close to or in contact with the head. A brain tumor can be present in any region within the brain, as depicted in our head model. The possible spatial locations of the 5 mm tumor are depicted in Fig. 9, and

comprehensive simulations were carried out for each specified position. As reported in Table II, the proposed antenna exhibits repeatable reflection-coefficient changes across all tumor positions, confirming stable sensitivity to malignant tissue despite the naturally small frequency displacement. Thus, the antenna maintains reliable detection capability in line with previously reported biomedical UWB antenna designs. Additionally, the SAR value is within the safe range for treatment planning, which is less than 1.6 W/Kg. Table III summarizes the analysis of the entire setup model respect to 7.79 GHz frequency, indicates deviation of current density and electric field intensity, return loss, VSWR, and directivity in the presence of an abnormal cell also significant variation of gain.

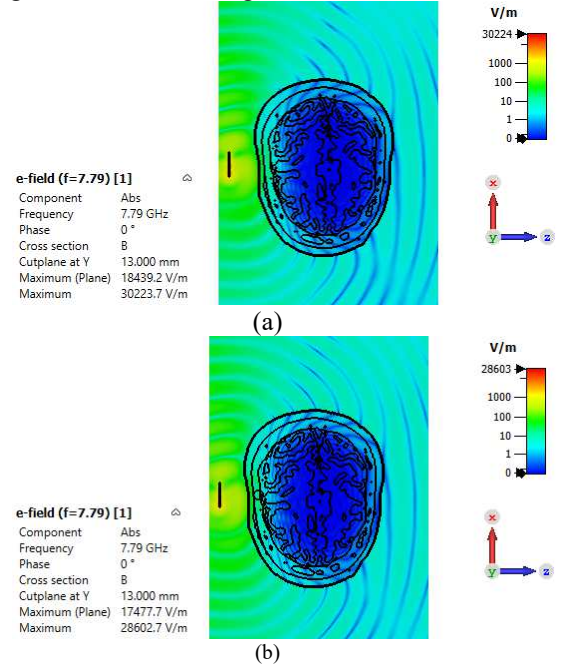


Fig. 6. E-field analysis at 7.79 GHz frequency (a) without tumor and (b) with tumor.

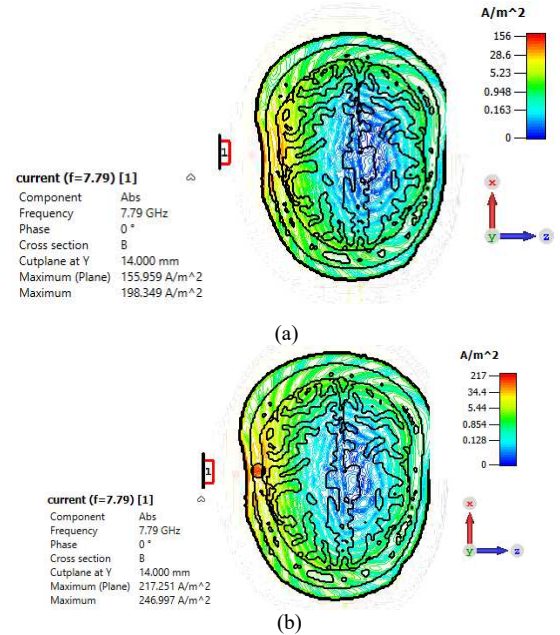


Fig. 7. Current density analysis at 7.79 GHz frequency (a) without tumor and (b) with tumor.

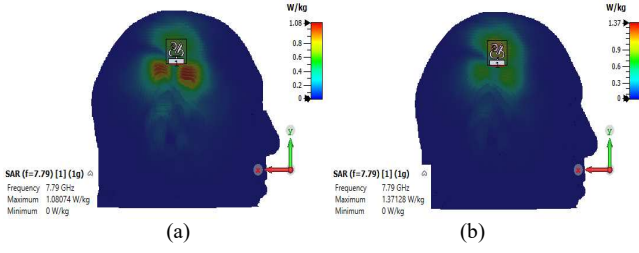


Fig. 8. SAR analysis at 7.79 GHz (a) without tumor and (b) with tumor.

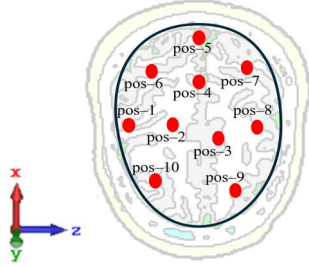


Fig. 9. Consideration of tumor position in human head model.

TABLE II RESULTS FOR TUMOR POSITION VARIATION AT 7.79 GHz

Position	Co-ordinate	Return loss (dB)	SAR (W/Kg) (1gm)	Depth (mm)
pos-1	10, 14.32, 34.35	-15.98	1.1003	14.35
pos-2	9, 12, 72	-15.992	0.9789	52
pos-3	7, 10, 97	-16.019	0.9743	77
pos-4	35, 16, 88	-15.992	0.9789	68
pos-5	68, 11, 99	-15.963	0.8736	79
pos-6	50, 13, 60	-15.988	0.8829	40
pos-7	48, 9, 120	-15.901	0.8723	100
pos-8	0, 8, 135	-16.191	0.8647	115
pos-8	-35, 12.37, 118	-16.126	0.8264	98
pos-9	-20, 10.41, 57	-15.992	0.9789	37

TABLE III PARAMETRIC COMPARISON FOR BRAIN PHANTOM

Parameter	In free space	Without tumor	With tumor
Resonance (GHz)	8.34	7.79	7.87
SAR for 1gm (W/Kg)	—	1.08	1.37
SAR for 10gm (W/Kg)	—	0.3594	0.5049
Electric field intensity (V/m)	84223.7	30225	28602.7
Current density (A/m ²)	47.88	198.35	246.96
Return loss (dB)	-40.73	-13.66	-15.24

TABLE IV COMPARISON WITH RELEVANT WORKS

Ref. no.	Size (L×W×h) mm ³	Substrate material	Operating freq. range (GHz)	BW (GHz)	Reflection coefficient (dB)	Centre frequency (GHz)	Gain (dB)	Max. SAR for 1g (W/kg)
[5]	57.6×57.6×3.4	—	2.24–2.56	0.32	-38.5571	2.4	5.5	0.1621911
[7]	60×70×1.5	FR4-epoxy	1.22–3.45	2.23	-45	1.8, 2.5	≈6	0.0316 (10gm)
[8]	56×37×1.6	FR4	1.54–2.75	1.21	-45	1.6, 2.2	>3.5	0.26
[9]	43×40×8.8	FR4	—	—	-50.55	2.45	—	0.84
[11]	31.68×31.01×0.75	Rogers R03003	—	—	—	7.3	—	—
[12]	40.43×38.78×0.018	Rogers RT5880	—	—	-57.98	2.45	—	0.57
[13]	44×30×1.6	FR4 epoxy	≈12.1–13.25	1.15	-20.023	12.454	—	—
[14]	60.46×78.73×1.7	FR4	1.5–3.0	1.5	-30.76	2.3	—	0.000635
[15]	36×32×1.6	FR4	2.307–2.587	0.28	≈-52	2.45	1.43	—
This Work	25×25×0.79	Rogers RT5880	6.68–8.82	2.14	-40.73	8.34	4	1.37

VSWR	1.02	1.52	1.42
Directivity (dBi)	4.568	1.82	2.58

IV. COMPARATIVE ANALYSIS

The proposed antenna features a compact design with a modified S-shaped patch, parasitic elements, and a partially truncated ground plane, enhancing bandwidth, gain, and radiation performance. Compared to existing antennas Table IV, it achieves significant size reduction, improved bandwidth, and superior radiation characteristics, addressing limitations such as low gain and narrow bandwidth. Designs such as [5], [15], and [14] offer bandwidths of 0.28 - 0.32 GHz, which restricts resolution and reduces the ability to detect small tumors or achieve accurate imaging. In contrast, the proposed design attains a broader operational bandwidth, higher radiation efficiency, and stable impedance characteristics, enabling more precise detection and improved signal penetration in heterogeneous tissue environments. Additionally, the antenna maintains a low-profile structure while ensuring consistent performance across the desired frequency range, overcoming challenges observed in previously reported designs.

V. CONCLUSION

This work introduces a miniaturized UWB microstrip antenna engineered for high-fidelity electromagnetic interaction with multilayer cranial tissues. The integration of an S-shaped radiator, parasitic strip, and hybrid ground topology enables wideband impedance stability and improved radiation robustness across 6.68–8.82 GHz. Full-wave simulations using a voxel-based human head model verified consistent sensitivity to dielectric perturbations induced by a 5 mm malignant inclusion, evidenced by measurable resonance drift, S-parameter variations, and field redistribution. The SAR distribution remained compliant with FCC exposure limits, confirming the electromagnetic safety of the design. The antenna's electrical stability, compact footprint, and reliable differential response indicate strong suitability for next-generation WBAN-based biomedical devices, enabling early cancer detection and personalized treatment in clinical practice.

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